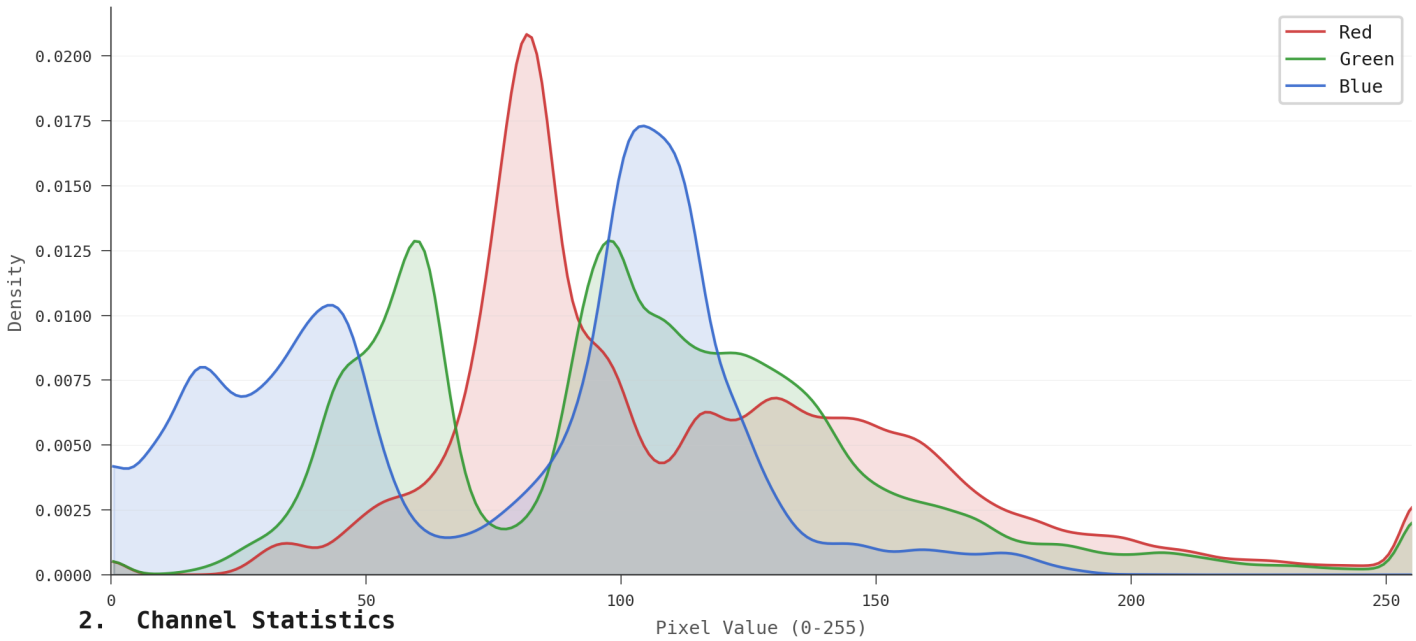
Reference Data Sheet: KODIM03

Subject: Neon Bahamas Hats on Fence

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	111.6838	44.0323	1938.8418	0.5779	0.8510	0	255
Green	101.9713	44.6306	1991.8898	0.6233	0.6694	0	255
Blue	76.0347	42.5095	1807.0590	-1.0205	-0.0510	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.7185	0.2890
G	0.7185	1.0000	0.5534
B	0.2890	0.5534	1.0000

Pairwise Pearson Correlation Coefficients

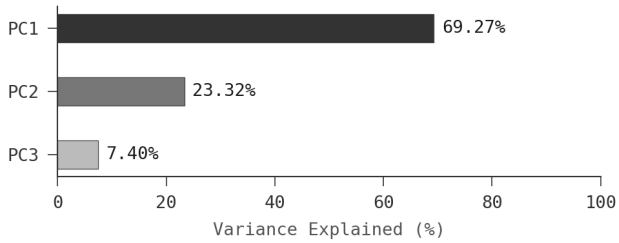
r(R-G)	=	**0.718536**
r(R-B)	=	**0.288997**
r(G-B)	=	**0.553408**
Avg |r|	=	**0.520314**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1938.8467} & 1412.0611 & 540.9428 \\
1412.0611 & \mathbf{1991.8949} & 1049.9428 \\
540.9428 & 1049.9428 & \mathbf{1807.0636}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM03

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5843	**-0.6635**	-0.4672	3974.7495
PC2	-0.5749	-0.0679	0.8154	1338.2911
PC3	-0.5728	0.7451	-0.3418	424.7647

6. Derived Metrics

Condition Number

9.36

(l1/l3) PC1/PC2: 2.97 | PC2/PC3: 3.15

Blue Channel Independence

52.0%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6635 | Green dominant

Classification: Three-Dimensional (PC1 = 69.27%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.983370	0.968858	0.976114
Green	0.984021	0.969870	0.976945
Blue	0.985672	0.970529	0.978101

8. Average Local Variance (3x3)

Channel	Value
Red	75.4177
Green	73.9969
Blue	58.5170

9. Redundancy Profile

Classification: Three-Dimensional | Condition Number: 9.36 | PC1: 69.27%

Correlation: R–G pair leads at r = 0.718536 | G–B: 0.553408 | R–B: 0.288997

Blue Channel: 52.0% independent | 3D color; major blue variance

Spatial: Avg autocorrelation: 0.977053 | Avg local variance: 69.3105

Eigenvector: Green dominant | G (-0.6635) leads | R (-0.5843), B (-0.4672)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."